

INTEGRATED BATTERY INTELLIGENCE SYSTEM

Internship Domain

Embedded Systems & IoT

Technologies Used

ESP32, Embedded C/C++, Wokwi, Blynk IoT, LCD HMI

INTERNSHIP PROJECT REPORT

Submitted by

SUDANAPALLI SRINIVAS

B.Tech. – Electronics and Communication Engineering

Vignan's Foundation for Science, Technology and Research (VFSTR)

TABLE OF CONTENTS

- 1.Cover Page
- 2.Table of Contents
- 3.Abstract
- 4.System Architecture
- 5.Workflow / Block Diagram
- 6.Implementation of the Six Internship Tasks
- 7.Hardware and Software Implementation
- 8.Blynk IoT Dashboard
- 9.Testing and Validation
- 10.Results and Discussion
- 11.Conclusion
- 12.Future Scope
- 13.Project Links

ABSTRACT

This internship project presents an Integrated Battery Intelligence System designed to monitor, analyze, protect, and remotely supervise a simulated four-cell lithium battery pack. The system integrates battery monitoring, event-driven safety protection, embedded HMI, fault-tolerant runtime management, cloud telemetry, and an IoT-based executive dashboard into a single unified platform.

An ESP32 microcontroller is used as the central controller to acquire the four cell voltages through analog inputs and perform real-time battery analytics. The system calculates individual cell voltages, pack voltage, average voltage, voltage imbalance, and identifies the weakest and strongest cells. Based on these parameters, the battery is classified into different health conditions including Healthy, Minor Imbalance, Critical Imbalance, and Pack Failure.

The protection subsystem continuously monitors abnormal conditions such as weak-cell voltage, overvoltage, sensor faults, rapid voltage fluctuations, frozen ADC conditions, and relay mismatches. A non-blocking, `millis()`-based event-driven architecture is used to manage protection actions, recovery, relay control, buzzer alerts, and system-state transitions.

A 16×2 LCD provides local diagnostic information, while Blynk IoT provides cloud-based telemetry and an executive dashboard for monitoring battery measurements, system status, faults, risk level, and operator recommendations. The complete system was implemented and tested in Wokwi, with nine fault and operating conditions validated through both the embedded interface and cloud dashboard.

The project demonstrates how multiple embedded-system and IoT functionalities can be integrated into a single scalable battery monitoring and protection platform suitable for intelligent battery-management applications.

INTRODUCTION

Battery-powered systems require continuous monitoring and reliable protection to ensure safe and efficient operation. In a multi-cell battery pack, differences in individual cell voltage can lead to imbalance, reduced performance, and potentially unsafe operating conditions. Therefore, an intelligent monitoring system must not only measure battery parameters but also analyze their behaviour, detect abnormal conditions, and respond appropriately.

This project focuses on developing a unified Embedded and IoT-based Battery Intelligence System using an ESP32 microcontroller. A simulated four-cell lithium battery pack is used as the input source, with potentiometers in the Wokwi environment representing individual cell voltage conditions. The system processes these inputs in real time and provides battery analytics, health classification, safety protection, fault detection, local diagnostics, and cloud-based monitoring.

The project combines six major functional areas into a single integrated system: battery intelligence, event-driven safety protection, embedded HMI and diagnostics, fault-tolerant runtime management, cloud telemetry, and an executive IoT dashboard. Instead of implementing these functions as separate projects, all modules operate together through a common embedded architecture.

The system uses a non-blocking `millis()`-based approach for timing and event management. This allows battery monitoring, fault detection, LCD updates, output control, and cloud communication to operate without blocking the main system execution. The design also includes system-state management, recovery mechanisms, relay protection, fault logging, and network reconnection handling.

The complete implementation was developed and tested using Wokwi and Blynk IoT. Multiple operating and fault conditions were tested to verify the behaviour of the integrated system. The final system demonstrates the practical integration of embedded intelligence and IoT technologies for battery monitoring and protection applications.

PROJECT OBJECTIVES

The main objective of this project is to develop a single integrated battery intelligence and IoT monitoring system that combines real-time battery analysis, safety protection, fault management, embedded diagnostics, cloud telemetry, and dashboard-based monitoring.

The specific objectives are:

1. To monitor the voltage of a simulated four-cell lithium battery pack using an ESP32.
2. To calculate important battery parameters such as individual cell voltage, pack voltage, average voltage, voltage imbalance, weakest cell, and strongest cell.
3. To classify battery health into Healthy, Minor Imbalance, Critical Imbalance, and Pack Failure conditions.
4. To implement an event-driven safety protection system for detecting weak-cell voltage, overvoltage, sensor anomalies, rapid voltage fluctuations, and other system faults.
5. To control the relay, buzzer, and status LEDs according to the detected system condition.
6. To develop an LCD-based Human-Machine Interface (HMI) for displaying battery measurements, analytics, system state, and fault diagnostics.
7. To implement a fault-tolerant runtime system with operational states such as Normal, Degraded, Failsafe, and Shutdown.
8. To implement non-blocking `millis()`-based timing for reliable and responsive system operation.
9. To provide cloud-based battery telemetry using Blynk IoT with network reconnection and event-driven data transmission.
10. To develop an executive battery dashboard displaying live measurements, battery health, system status, faults, risk information, and operational recommendations.
11. To validate the complete integrated system through multiple Wokwi and Blynk test scenarios.
12. To demonstrate that all six internship tasks can operate together as one unified embedded and IoT system.

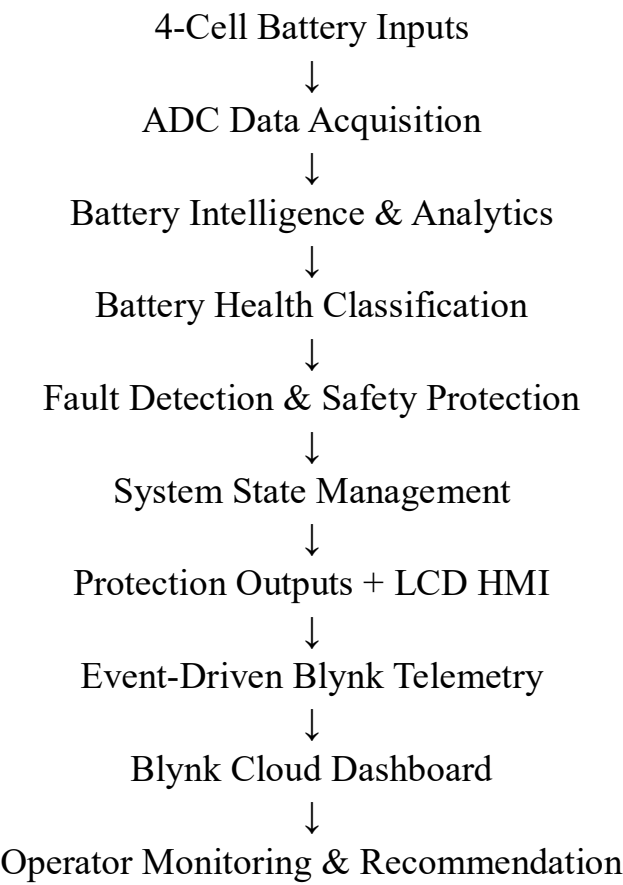
SYSTEM ARCHITECTURE

Module Integration

The six internship tasks are integrated as follows:

Module	Function
Battery Intelligence Engine	Battery measurement and analytics
Safety Protection Kernel	Real-time fault detection and protection
Embedded HMI & Diagnostics	LCD-based system visualization
Fault-Tolerant Runtime	Fault handling and system-state management
Cloud Telemetry Architecture	Blynk communication and telemetry
Executive Dashboard	Remote monitoring and operational analytics

WORKFLOW / BLOCK DIAGRAM



IMPLEMENTATION OF THE SIX INTERNSHIP TASKS

6.1 Adaptive Multi-Cell Battery Intelligence Engine

The Battery Intelligence Engine forms the core measurement and analytics layer of the system. The ESP32 continuously acquires the voltage values of the four simulated battery cells through analog inputs.

The system calculates the voltage of each cell and derives important pack-level parameters including total pack voltage, average cell voltage, minimum and maximum cell voltage, voltage imbalance percentage, weakest cell, and strongest cell.

Based on the calculated imbalance and cell conditions, the system classifies battery health into four categories:

- Healthy
- Minor Imbalance
- Critical Imbalance
- Pack Failure

This module provides the processed battery information required by the safety, HMI, telemetry, and dashboard modules.

6.2 Event-Driven Safety Protection Kernel

The Safety Protection Kernel continuously evaluates the battery measurements and identifies abnormal operating conditions. The implemented system detects:

- Weak cell voltage
- Cell overvoltage
- Sensor fault
- Rapid voltage fluctuation
- Frozen ADC condition
- Relay mismatch

Fault confirmation and recovery are handled using non-blocking timing. The system uses millis()-based event-driven logic without delay(), allowing monitoring and protection functions to continue operating simultaneously.

Depending on the detected condition, the system controls the relay, buzzer, and red, yellow, and green status LEDs to provide immediate protection and indication.

6.3 Intelligent Embedded HMI & Diagnostic Interface

A 16×2 I2C LCD is used as the local Human-Machine Interface. The display automatically cycles through multiple information screens when the system is operating normally.

The HMI displays:

- Pack voltage and average voltage
- Individual cell voltages
- Voltage imbalance
- Weakest and strongest cells
- System state
- Fault information
- Fault count

When a critical fault occurs, the system gives priority to the fault diagnostic screen instead of continuing the normal screen rotation.

6.4 Fault-Tolerant Embedded Runtime System

The runtime subsystem manages the overall operating state of the battery system. Based on battery health and detected faults, the system transitions between:

NORMAL → DEGRADED → FAILSAFE → SHUTDOWN

The runtime system also maintains fault information, fault counters, recovery timing, and system-state transitions. Fault conditions are isolated and handled according to their severity while allowing normal monitoring functions to continue whenever possible.

6.5 Intelligent Cloud Telemetry Architecture

The cloud telemetry module connects the ESP32 to the Blynk IoT platform through Wi-Fi.

The system transmits battery measurements and diagnostic information through Blynk virtual datastreams. Telemetry is triggered by important changes such as

battery-data changes, system-state transitions, fault events, and reconnection events rather than relying only on continuous fixed-interval transmission.

The system also includes Wi-Fi and Blynk reconnection handling so that the embedded battery monitoring and protection functions continue operating even when cloud connectivity is temporarily unavailable.

6.6 Executive Battery Intelligence Dashboard

The Blynk dashboard provides a centralized interface for monitoring the integrated battery system.

The dashboard displays:

- Individual cell voltages
- Pack voltage
- Average voltage
- Voltage imbalance
- Battery health
- System state
- Active fault
- Fault count
- Relay status
- Risk level
- Operator recommendation
- Battery voltage visualization

The dashboard allows the operator to observe the system's current condition and identify abnormal operating conditions through structured telemetry and status information.

HARDWARE AND SOFTWARE IMPLEMENTATION

Component	Purpose
ESP32	Main processing and control unit
4 Potentiometers	Simulated four battery-cell voltage inputs
16×2 I2C LCD	Local HMI and diagnostic display
Green LED	Normal operating indication
Yellow LED	Degraded operating indication
Red LED	Failsafe/Shutdown indication
Buzzer	Audible fault indication
Relay	Battery protection/cutoff control
Wi-Fi	Cloud communication

7.2 Software and Platforms

Software / Platform	Purpose
Arduino IDE / Wokwi	ESP32 programming and simulation
Embedded C/C++	System implementation
Blynk IoT	Cloud telemetry and dashboard
LiquidCrystal_I2C	LCD communication
Blynk Library	IoT communication
Wi-Fi Library	Network connectivity

7.3 Pin Configuration

Function	ESP32 Pin
Cell 1	GPIO 34
Cell 2	GPIO 35
Cell 3	GPIO 32
Cell 4	GPIO 33
Red LED	GPIO 2
Green LED	GPIO 4
Yellow LED	GPIO 5
Buzzer	GPIO 18
Relay	GPIO 19
LCD SDA	I2C
LCD SCL	I2C

BLYNK IOT DASHBOARD

The Blynk IoT dashboard is used for remote monitoring of the battery system. It displays:

- Cell voltages
- Pack voltage and average voltage
- Imbalance percentage
- Battery health
- System state and active fault
- Fault count and relay status
- Risk level and operator recommendation

TESTING AND VALIDATION

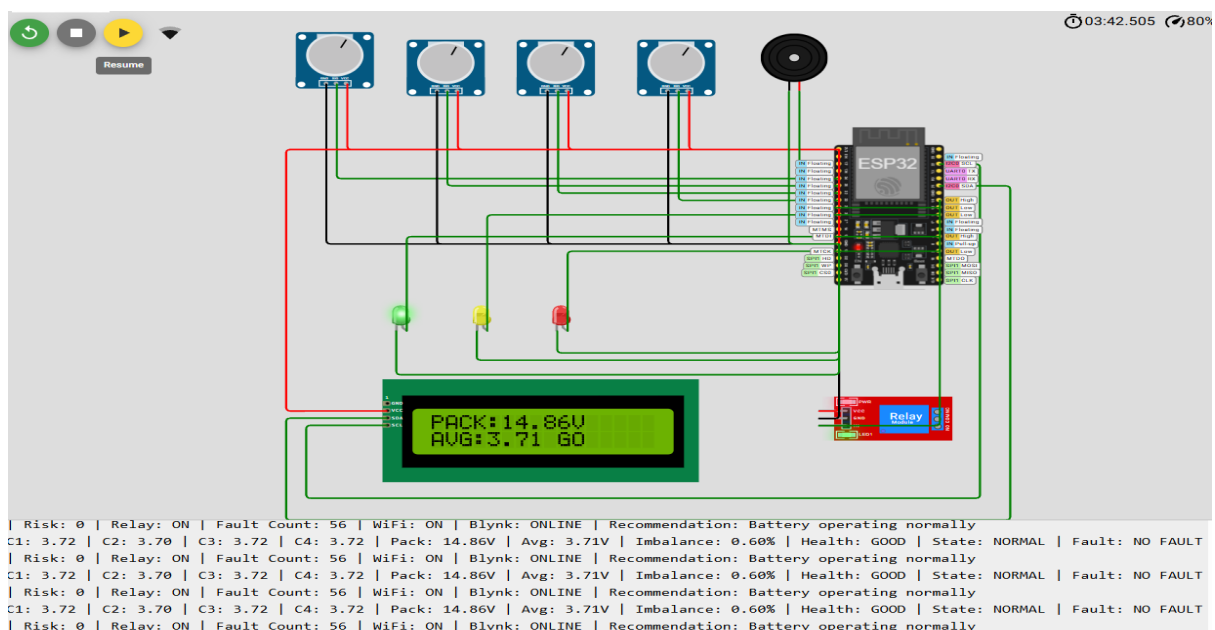
The integrated system was tested under nine different operating and fault conditions using Wokwi and the Blynk IoT dashboard.

No.	Test Condition	Expected State
1	Healthy Battery	NORMAL
2	Minor Imbalance	DEGRADED
3	Critical Imbalance	DEGRADED
4	Weak Cell	FAILSAFE
5	Overvoltage	FAILSAFE
6	Rapid Voltage Change	FAILSAFE
7	Sensor Fault	SHUTDOWN
8	Frozen ADC	SHUTDOWN
9	Recovery	NORMAL

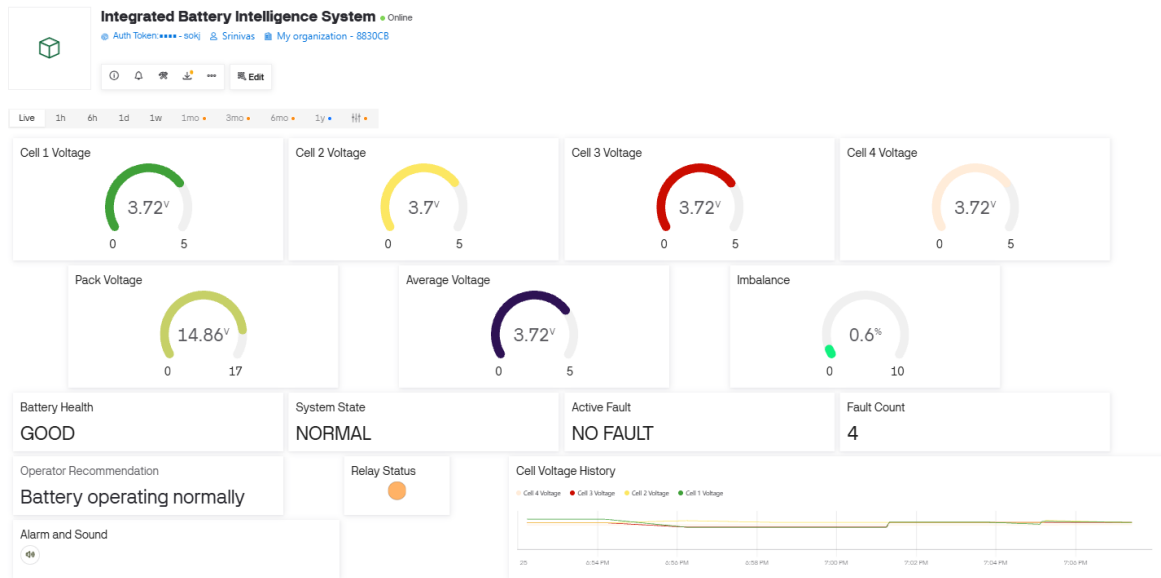
Each condition was verified through the Wokwi simulation, LCD/outputs, Serial Monitor, and Blynk dashboard.

9.1 Test Evidence

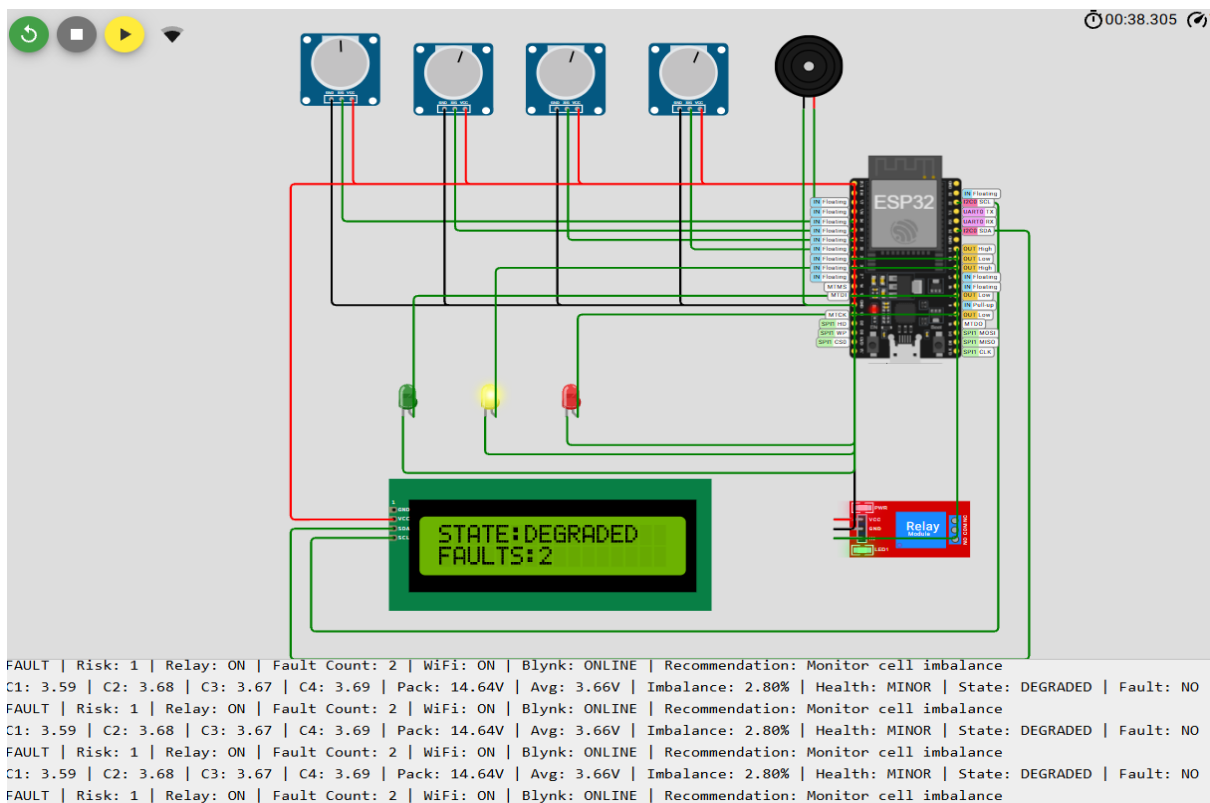
Test 1 – Healthy Battery, Wokwi



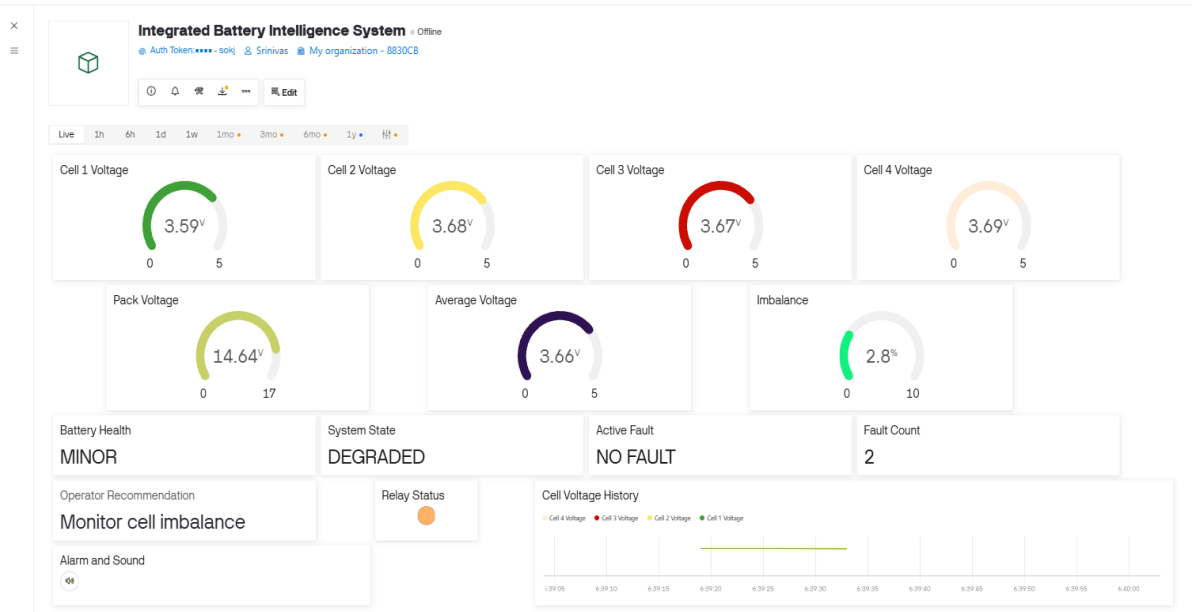
Blynk



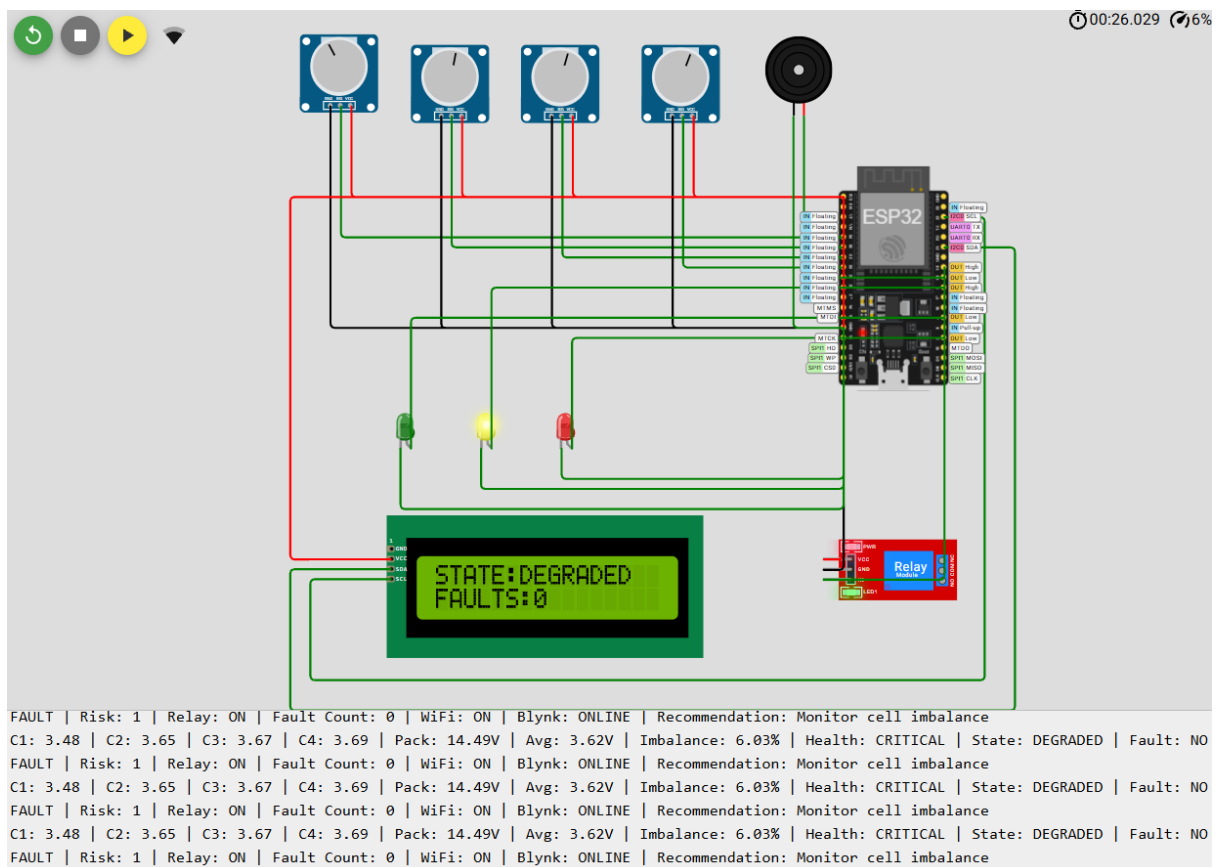
TEST 2, Minor imbalance



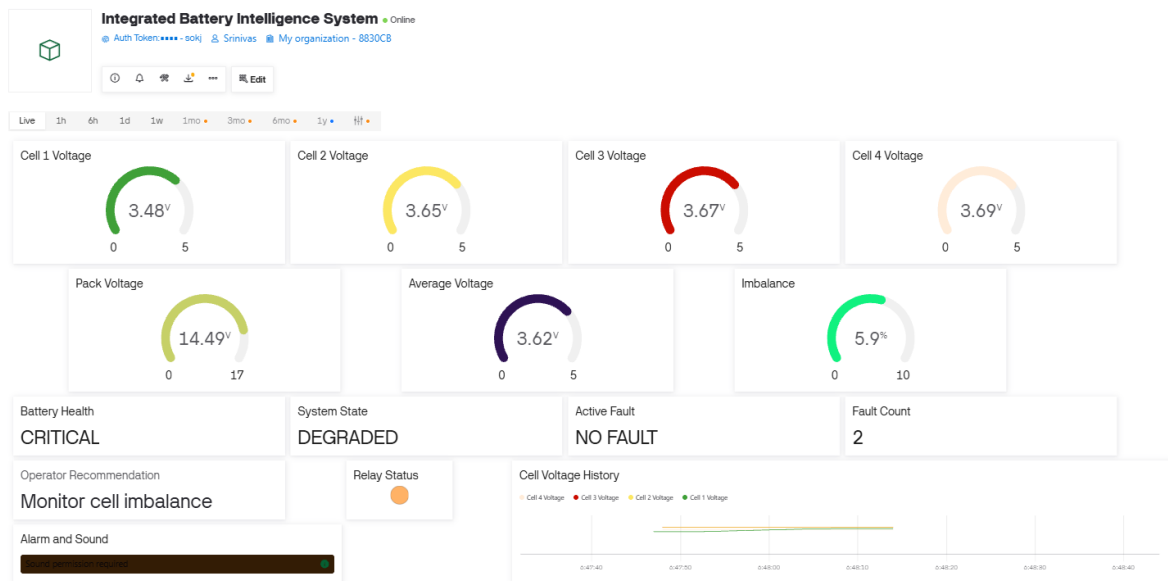
Blynk, minor imbalance



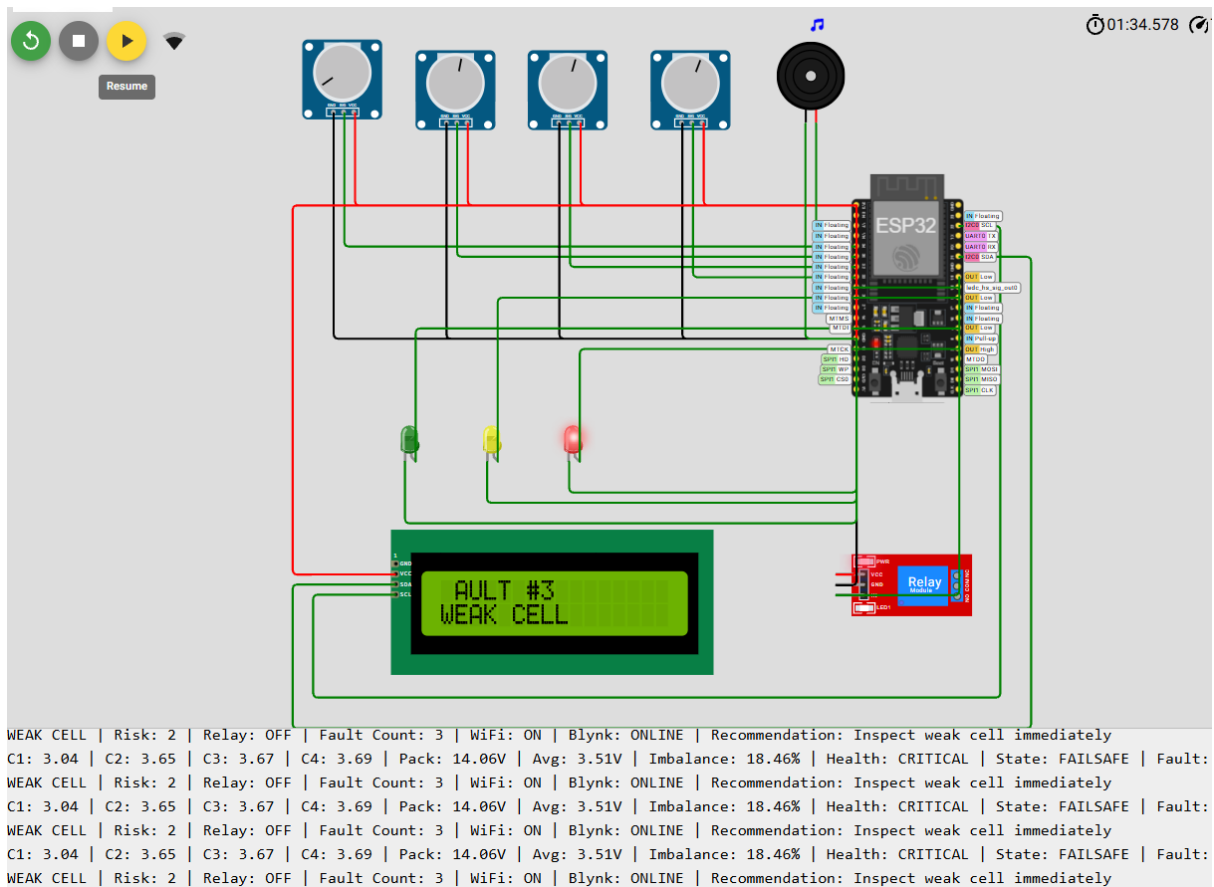
Test 3, critical imbalance, wokwi



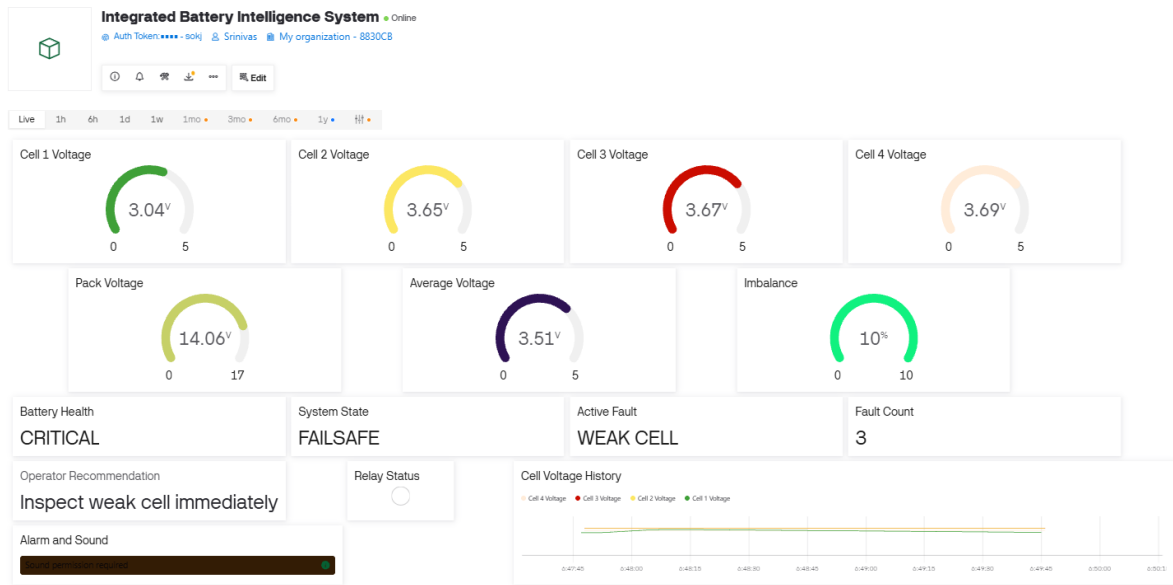
Blynk, critical imbalance



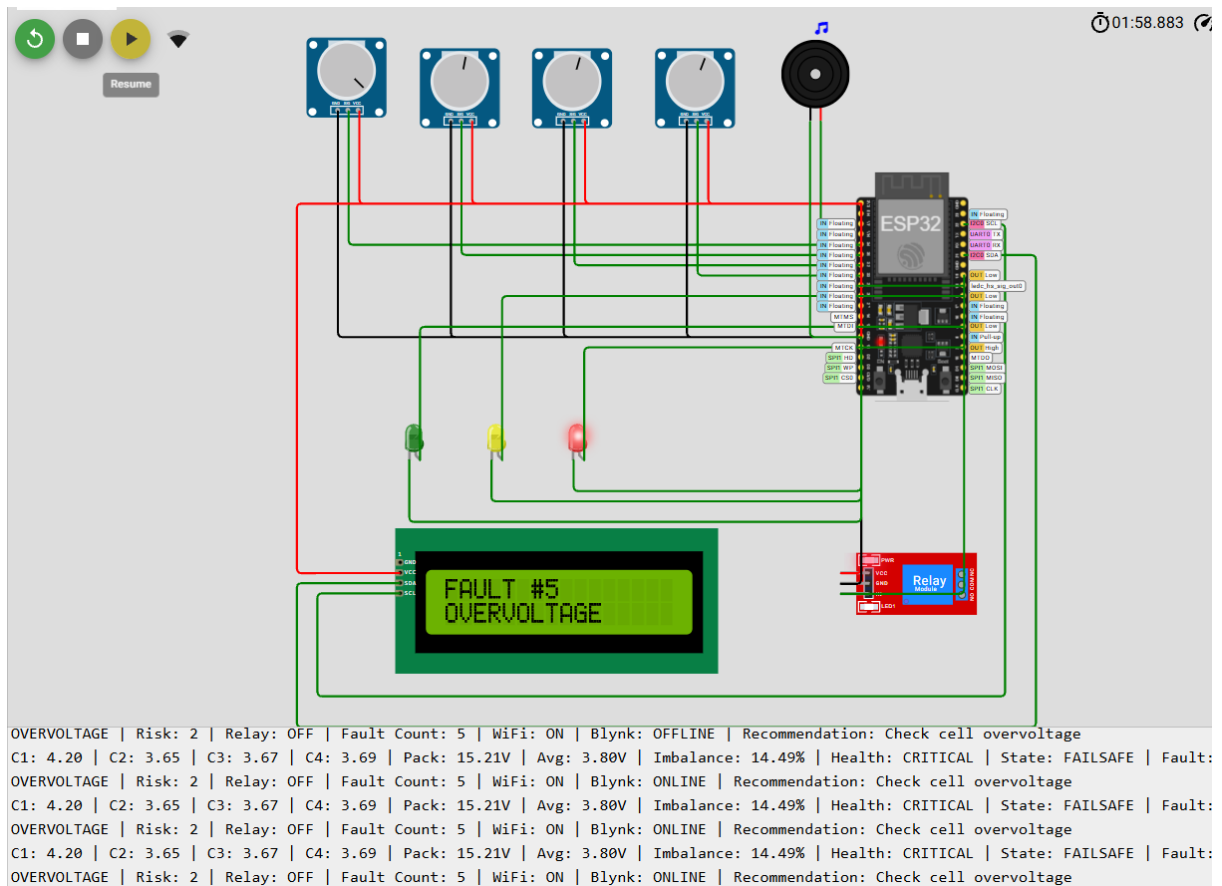
Test 4, Weak cell, Wokwi



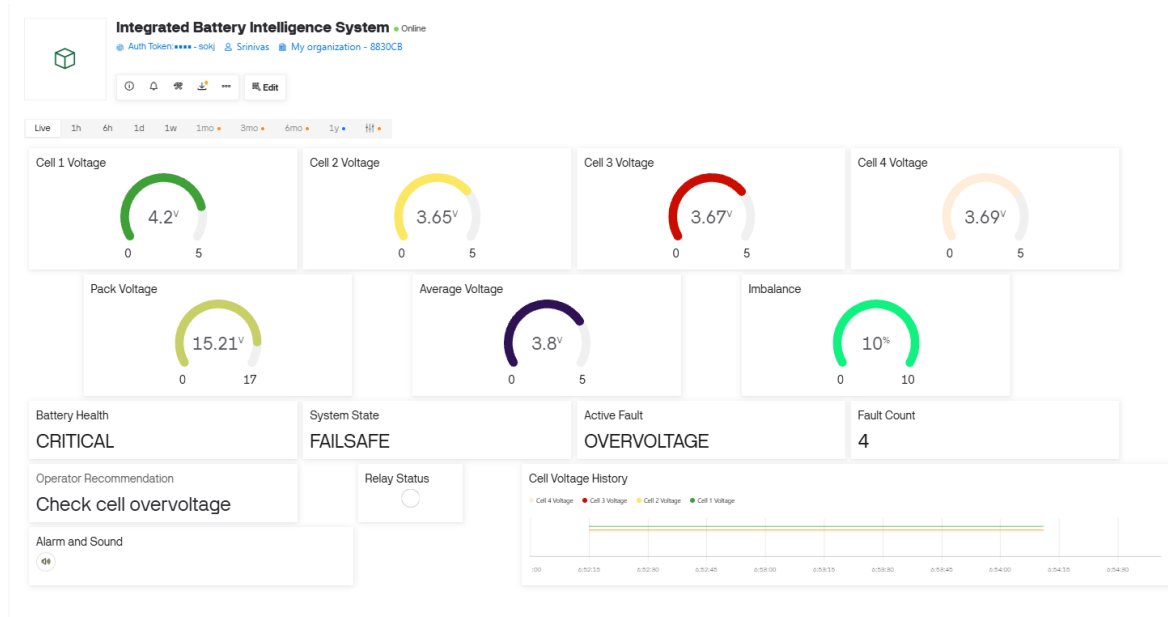
Blynk, weak cell



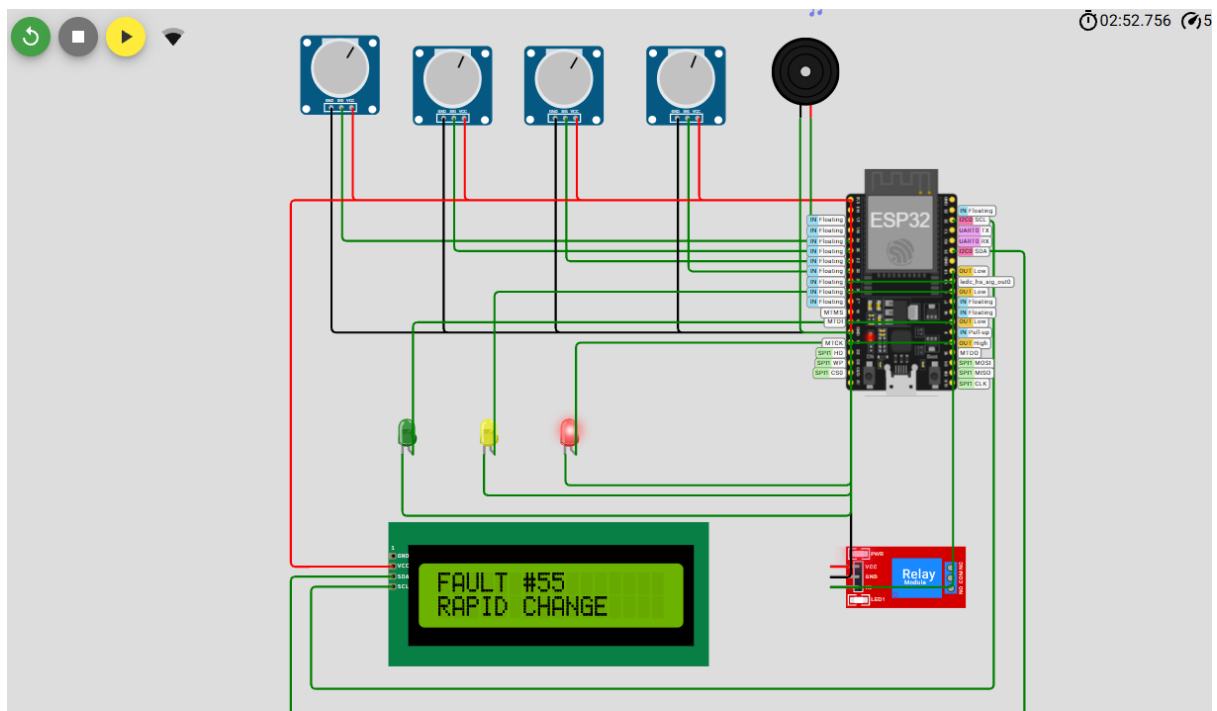
Test 5, over voltage, Wokwi



Blynk, over voltage

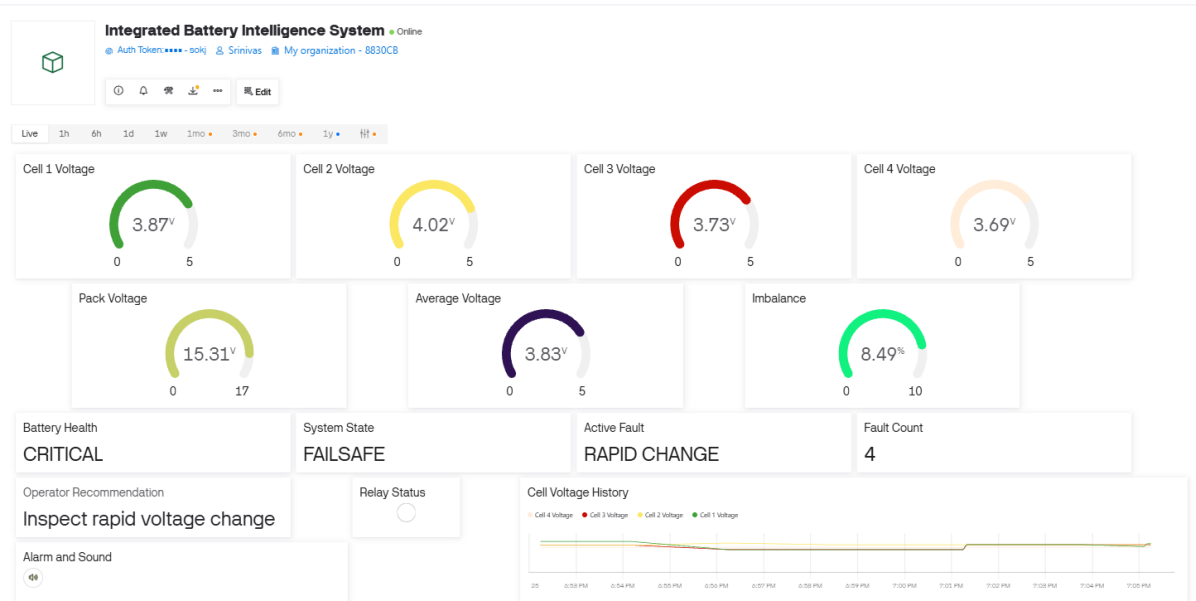


Test 6, Rapid voltage change

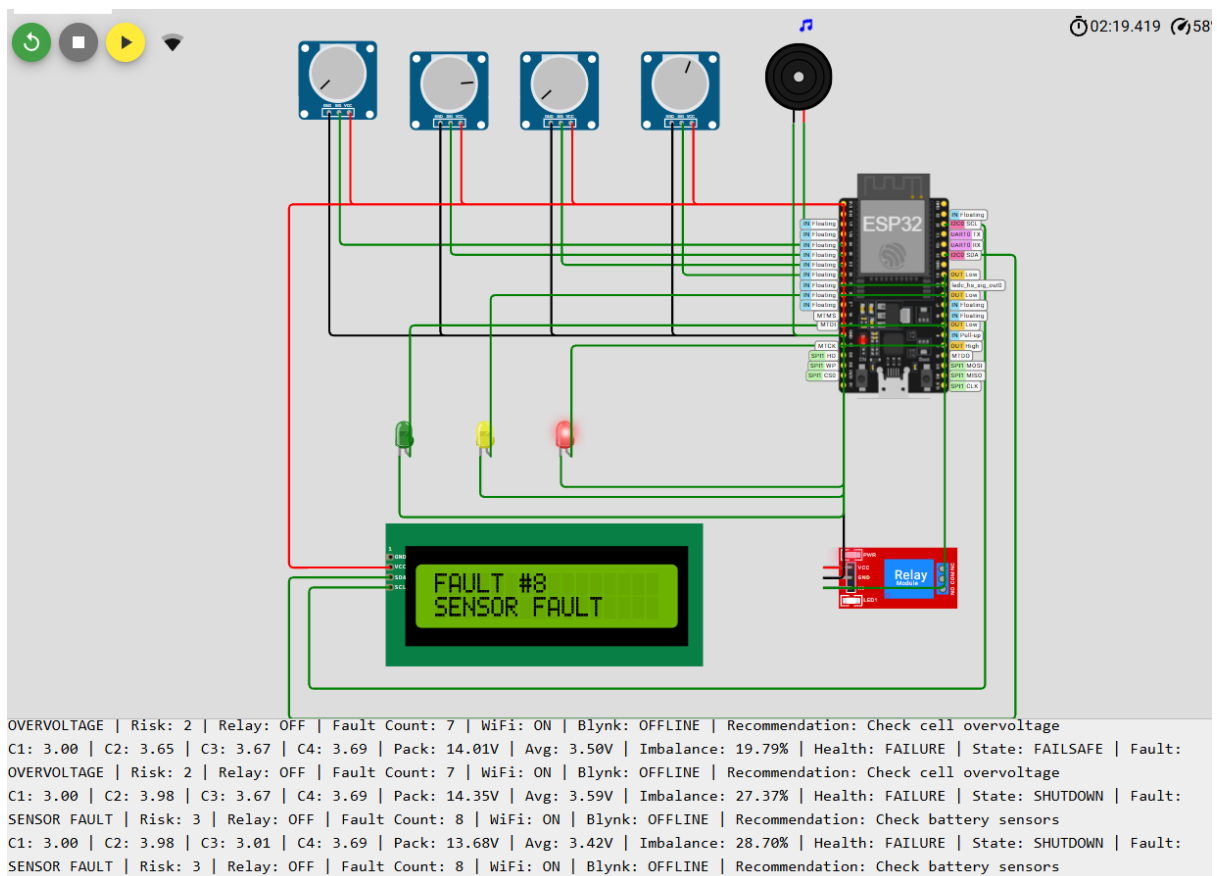


WEAK CELL | Risk: 2 | Relay: OFF | Fault Count: 54 | WiFi: ON | Blynk: ONLINE | Recommendation: Inspect weak cell immediately
C1: 3.74 | C2: 3.71 | C3: 3.01 | C4: 3.69 | Pack: 14.15V | Avg: 3.54V | Imbalance: 20.60% | Health: CRITICAL | State: FAILSAFE | Fault:
WEAK CELL | Risk: 2 | Relay: OFF | Fault Count: 54 | WiFi: ON | Blynk: ONLINE | Recommendation: Inspect weak cell immediately
C1: 3.74 | C2: 3.71 | C3: 3.73 | C4: 3.69 | Pack: 14.87V | Avg: 3.72V | Imbalance: 1.20% | Health: GOOD | State: FAILSAFE | Fault: RAPID
CHANGE | Risk: 2 | Relay: OFF | Fault Count: 55 | WiFi: ON | Blynk: ONLINE | Recommendation: Inspect rapid voltage change
C1: 3.74 | C2: 3.71 | C3: 3.73 | C4: 3.69 | Pack: 14.87V | Avg: 3.72V | Imbalance: 1.20% | Health: GOOD | State: FAILSAFE | Fault: RAPID
CHANGE | Risk: 2 | Relay: OFF | Fault Count: 55 | WiFi: ON | Blynk: ONLINE | Recommendation: Inspect rapid voltage change

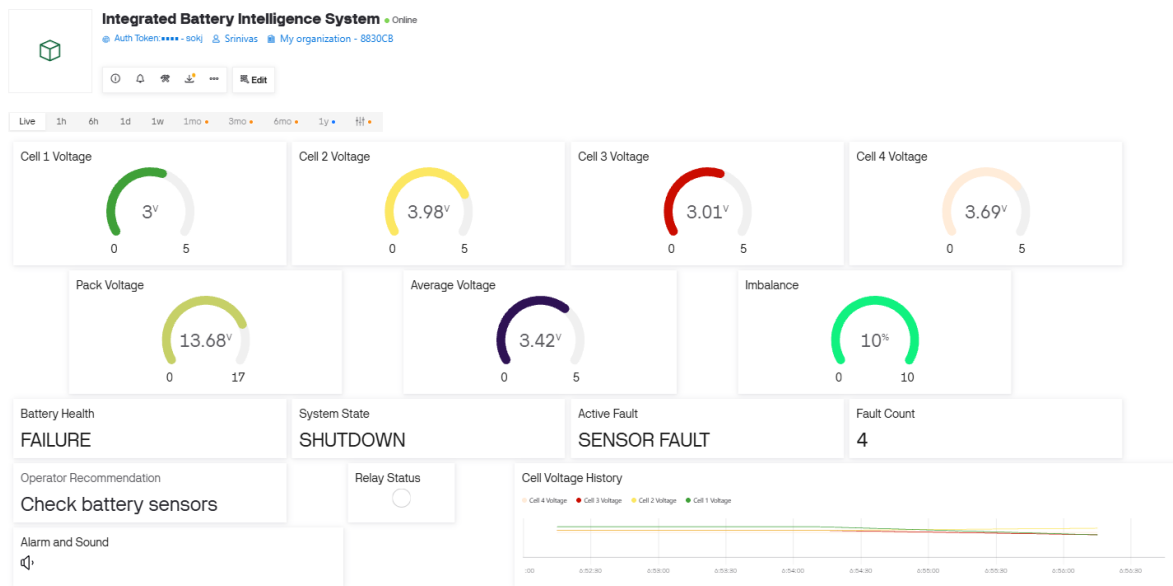
Blynk, Rapid voltage change



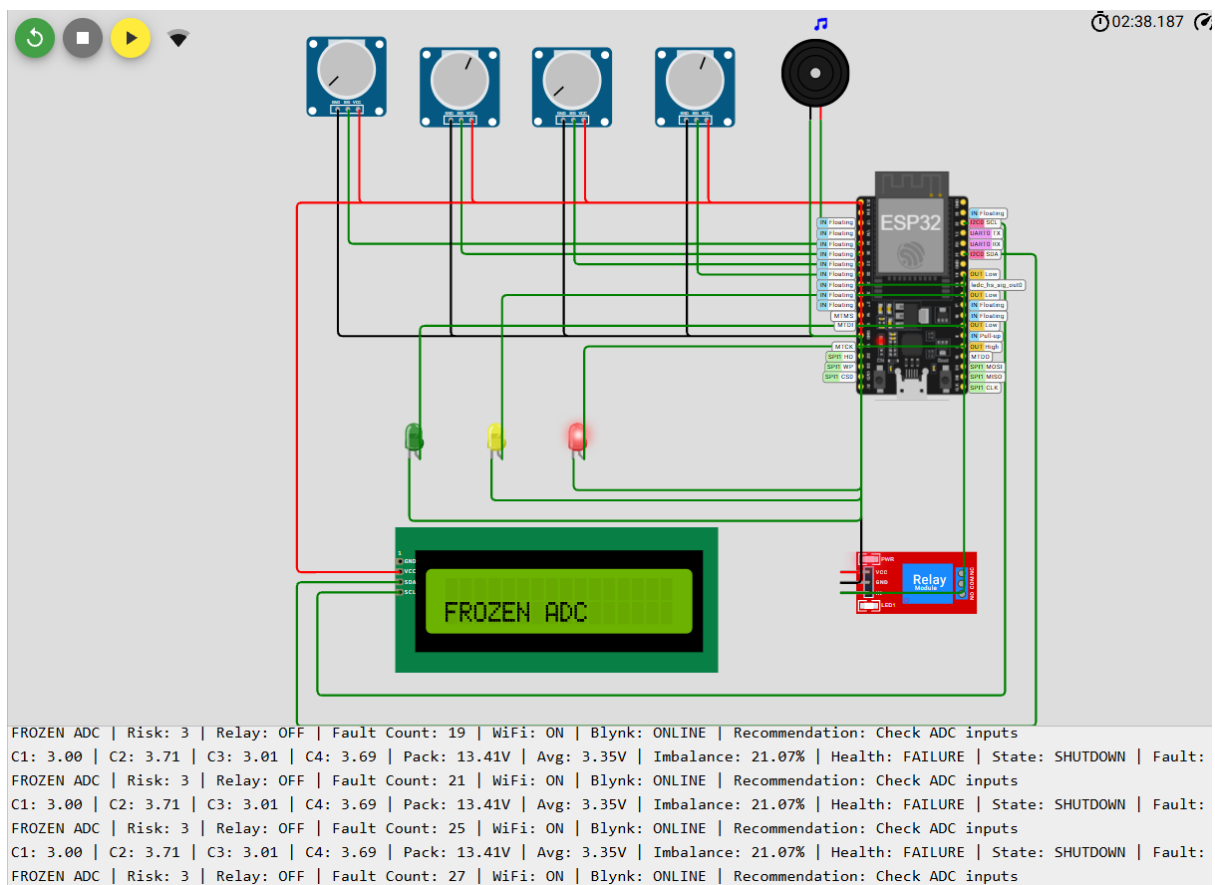
Test 7, sensor fault/shutdown, Wokwi



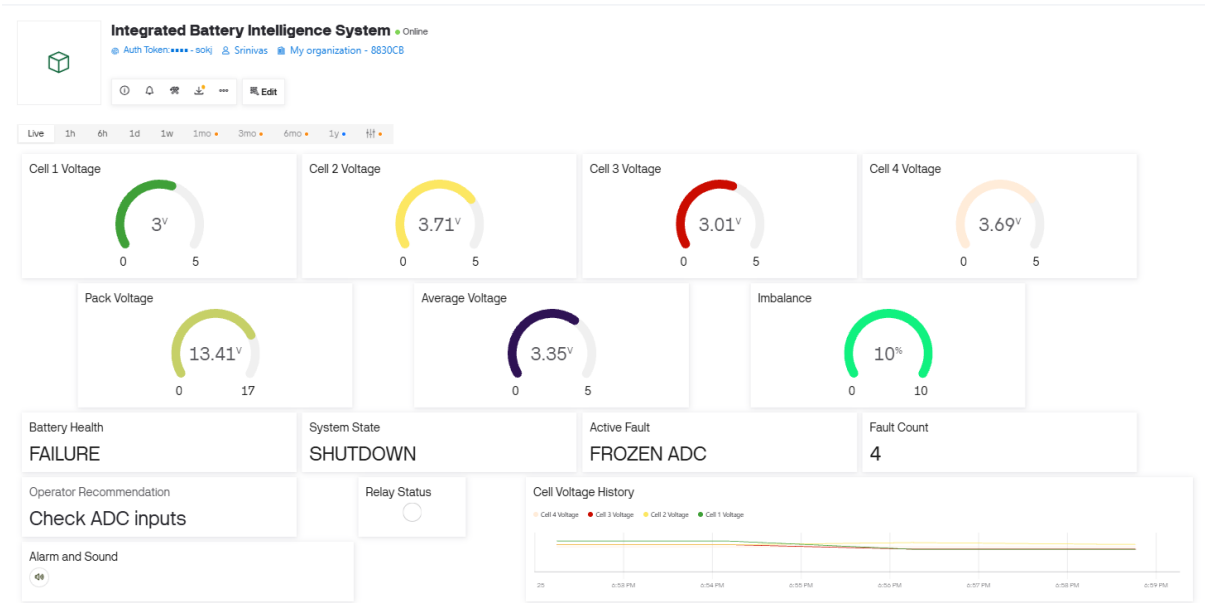
Blynk, sensor fault/shutdown



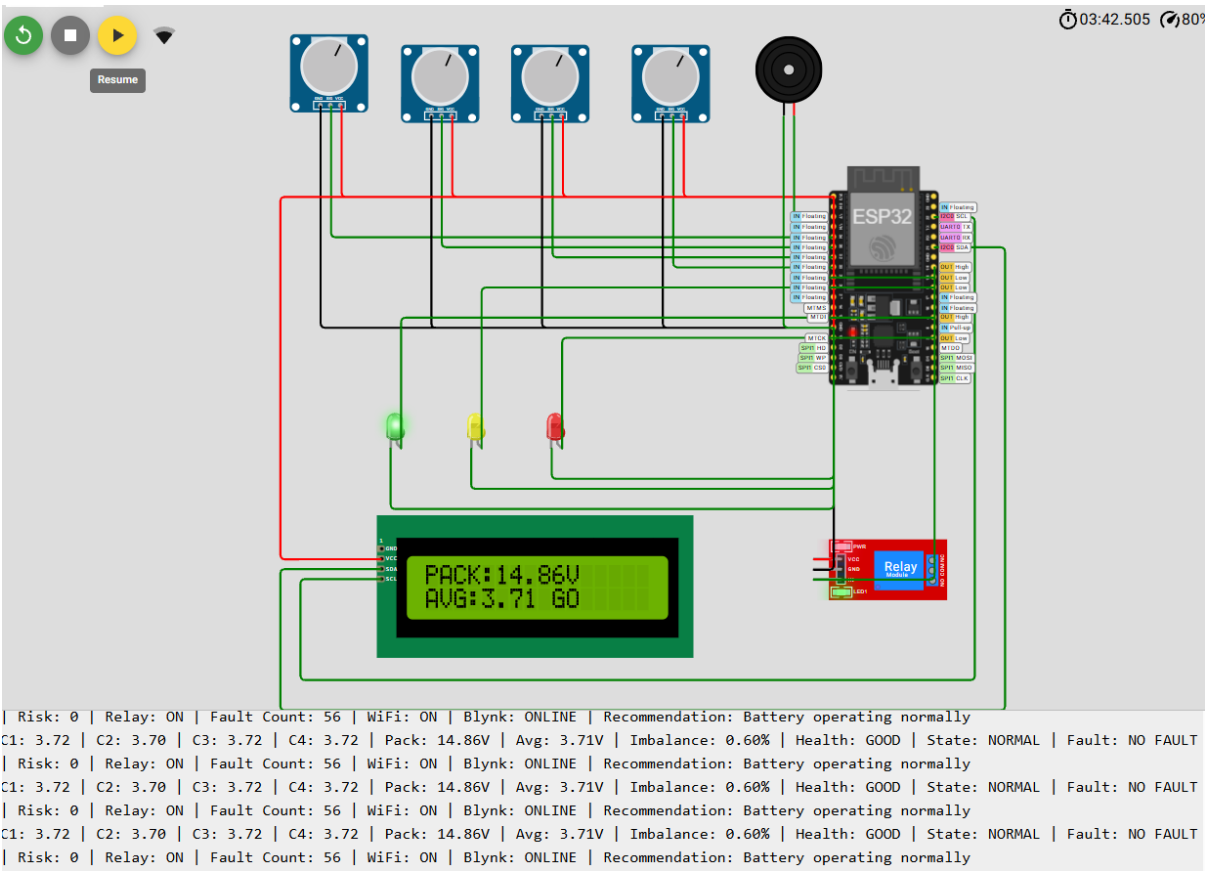
Test 8, Frozen ADC, Wokwi



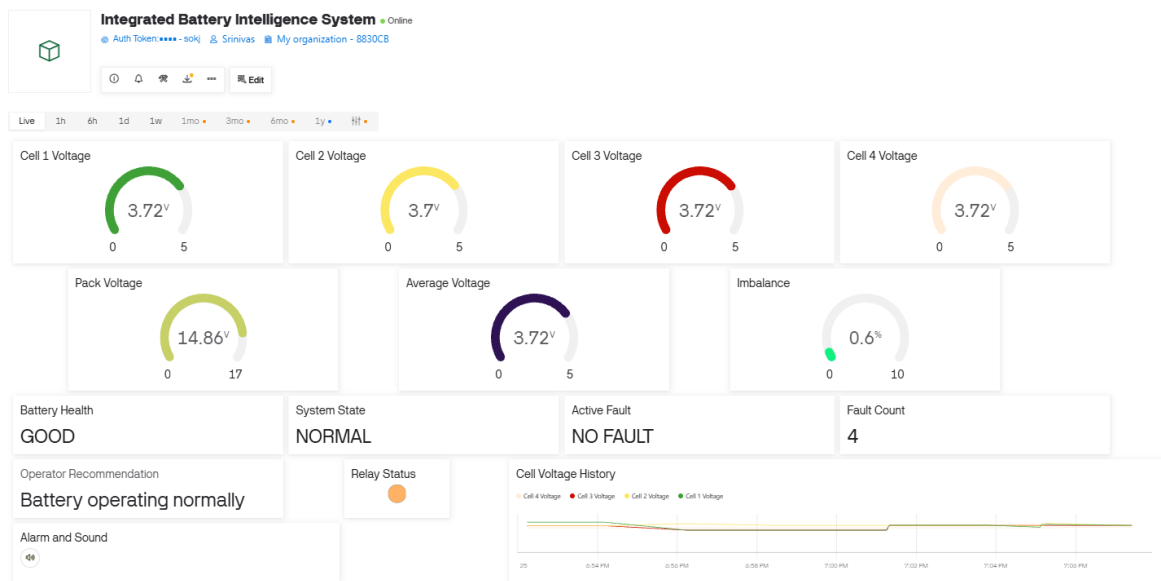
Blynk, Frozen ADC



Test 9, Recovery, Wokwi



Blynk, recovery



RESULTS AND DISCUSSION

The integrated battery intelligence system was successfully implemented and tested in the Wokwi simulation environment with Blynk IoT integration.

The system successfully demonstrated:

- Four-cell voltage monitoring and battery analytics.
- Battery health classification.
- Fault detection and protection.
- normal, degraded, failsafe, and shutdown states.
- LCD-based diagnostics.
- Relay, LED, and buzzer control.
- Blynk cloud monitoring and telemetry.
- Successful validation of all nine test conditions.

The test results confirm that the six internship tasks function together as one integrated battery monitoring, protection, diagnostic, and IoT system.

CONCLUSION

The **Integrated Battery Intelligence System** was successfully developed by combining all six internship tasks into a single ESP32-based embedded and IoT platform.

The system successfully performs real-time battery monitoring, analytics, fault detection, safety protection, LCD diagnostics, fault-tolerant state management, Blynk telemetry, and dashboard monitoring. The complete system was validated through nine test conditions using Wokwi and Blynk.

The project demonstrates the practical integration of Embedded Systems, IoT, battery monitoring, safety protection, and diagnostic technologies into a unified system

FUTURE SCOPE

The system can be further enhanced with:

- Real battery cells and sensors instead of simulation.
- State-of-Charge (SoC) and State-of-Health (SoH) estimation.
- Temperature monitoring and thermal protection.
- Battery charging and balancing control.
- Long-term cloud data analysis and predictive maintenance.
- Mobile notifications and advanced fault alerts.
- Integration with automotive Battery Management Systems (BMS).

PROJECT LINKS AND SUBMISSION DETAILS

Wokwi Simulation: <https://wokwi.com/projects/473061276088437761>

Blynk: <https://blynk.cloud/dashboard/706483/global/devices/215401/organization/706483/devices/2245426/dashboard>

GitHub: <https://github.com/srinivas79798/integrated-battery-intelligence-system>